

Find and Fix: Separable Equations and Initial Conditions

Directions

- Every problem shows work that is wrong somewhere. Find the error, then repair it.
 - Part (a) always wants a corrected value. Put it on the Answer line.
 - Part (b) always wants a sentence or two naming the error — not just “it is different”, but which rule was broken. Write it in the space above the Answer line.
 - Round only at the last step, to the number of decimal places asked for.
1. A student separates $\frac{dy}{dx} = \frac{x}{y}$ correctly as $y \, dy = x \, dx$, then integrates to get $y^2 = x^2 + C$.
- (a) Give the correct result of integrating both sides, in the form $\frac{1}{2}y^2 = G(x) + C$.
- (b) Name the error the student made.

Answer: _____

2. To solve $\frac{dy}{dx} = 6x^2y$ with $y(0) = 4$, a student writes $\ln|y| = 6x^3 + C$ and reports $y(1) = 4e^6 \approx 1613.715$.

- (a) Find the correct value of $y(1)$, to three decimal places.
- (b) Name the error the student made.

Answer: _____

3. For $\frac{dy}{dx} = \frac{y}{5}$ with $y(0) = 7$, a student integrates correctly to $\ln|y| = \frac{x}{5} + C$, then writes $y = e^{x/5} + C$. From $y(0) = 7$ they get $C = 6$ and report $y(10) = e^2 + 6 \approx 13.389$.

- (a) Find the correct value of $y(10)$, to three decimal places.
- (b) Name the error the student made.

Answer: _____

4. A student solves $\frac{dy}{dx} = \frac{4x}{y}$ with $y(0) = 3$ and $y > 0$. Their work reads: $y dy = 4x dx$, then $y^2 = 2x^2 + C$, then $C = 9$, and finally $y(2) \approx 4.123$.

- (a) Find the correct value of $y(2)$.
- (b) Name the error the student made, and say why the value of C they found is wrong as a consequence of it.

Answer: _____

5. A savings balance P , in dollars, satisfies $\frac{dP}{dt} = 0.06P$ with $P(0) = 2000$, where t is in years. A student writes $P = e^{0.06t} + 2000$ and reports $P(15) \approx 2002.46$.

- (a) Find the correct value of $P(15)$, in dollars, to two decimal places.
- (b) Name the error, and describe the quick sanity check on the story that should have caught it.

Answer: _____

6. A student looks at $\frac{dy}{dx} = x + xy$ and says the equation cannot be separated because the right side is a sum.

- (a) Separate the equation anyway, solve it with $y(0) = 2$, and give $y(1)$ to three decimal places.
- (b) Explain the test that decides whether a right side containing a sum is separable.

Answer: _____

7. Two students solve $\frac{dy}{dx} = \frac{\cos x}{2y}$ with $y(0) = 3$ and $y > 0$. One writes $y^2 = \sin x + 9$; the other writes $y^2 = \sin x + 3$ and reports $y(\frac{\pi}{2}) = 2$.

- (a) Find the correct value of y at $x = \frac{\pi}{2}$, to three decimal places.
- (b) Explain which of the two solutions is wrong and exactly what was done incorrectly.

Answer: _____

8. Here is a complete student solution to $\frac{dy}{dx} = \frac{2x}{y+1}$ with $y(0) = 3$ and $y > -1$. Exactly one line is wrong.

Line 1: $(y+1) dy = 2x dx$

Line 2: $\frac{(y+1)^2}{2} = x^2 + C$

Line 3: substituting $(0, 3)$ gives $C = 16$

Line 4: $y = -1 + \sqrt{2x^2 + 32}$, so $y(2) \approx 5.325$

(a) Give the corrected value of $y(2)$, to three decimal places.

(b) Identify the wrong line and explain the mistake in it.

Answer: _____